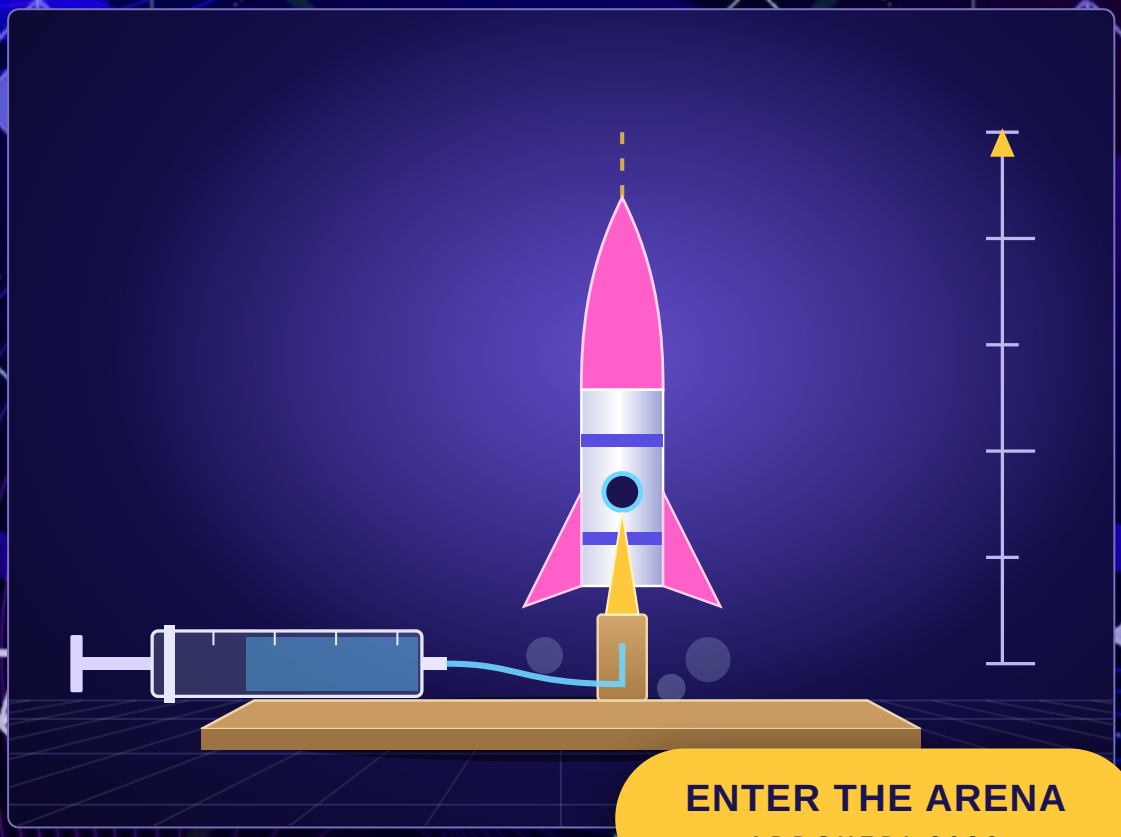
NIT ARUNACHAL PRADESH

ADDOVEDI

ENTER THE ARENA

SKYFORGE

PNEUMATIC ROCKET AVIATION CHALLENGE • RULEBOOK



ENTER THE ARENA

ADDOVEDI 2026

ORGANISED BY

Mechanical Engineering Student's Society (MESS)

National Institute of Technology Arunachal Pradesh

ADDOVEDI 2026 • Theme: Enter the Arena

1. Event Overview

SKYFORGE is the pneumatic rocket aviation challenge of ADDOVEDI 2026, organised by MESS, NIT Arunachal Pradesh. In this **two-round**, hands-on competition, participants learn the fundamentals of aerospace engineering by designing, building and launching a lightweight model rocket.

Every rocket is launched with the **standardised syringe-based pneumatic propulsion system** provided by the organisers. The objective is the **maximum valid flight height**, achieved within all safety, dimensional, structural and competition requirements.

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WHAT THE COMPETITION TESTS

AREA	WHAT PARTICIPANTS DEMONSTRATE
Aerodynamic design	A body and nose cone shaped to move cleanly through the air
Rocket stability	A rocket that flies straight and true after launch
Centre of gravity	Correct balance of mass along the rocket
Fin configuration	Fin number, shape and placement that keep the rocket stable
Weight distribution	Mass placed to support a stable, efficient flight
Lightweight construction	Minimum mass without losing strength
Drag reduction	Smooth surfaces and a streamlined profile
Structural integrity	A rocket that withstands launch without failing
Pneumatic propulsion	Working effectively with the standard syringe-based launcher
Engineering problem-solving	Practical design decisions and quick corrections

2. Team Formation

3	₹50	Open
MEMBERS PER TEAM	REGISTRATION FEE / MEMBER	NO LIMIT ON TEAMS

2.1 Each team must consist of **exactly 3 members**.

2.2 There is **no maximum limit** on the total number of teams that may register for the event.

2.3 The registration fee is **₹50 per member**.

2.4 Each participant may belong to only one team.

2.5 All team members must complete the registration process.

2.6 Accurate participant details must be provided during registration.

2.7 Any change in team composition requires approval from the organisers.

3. Component Allocation

The organisers will provide construction kits to a limited number of teams. Kits are allotted only to the **first 8 successfully registered teams**. Registration for the event itself remains open to all teams.

3.1 Component kits will be distributed strictly on a **first come, first served** basis.

3.2 A team becomes eligible for a component kit only after completing both registration and payment.

3.3 Only **8 organiser-provided kits** are available.

3.4 Once all 8 kits have been allocated, no further organiser-provided kits will be available.

3.5 Teams registering after the first 8 may still register and participate, subject to the event rules, but will not receive the organiser-provided construction kit.

3.6 The organisers will announce when all 8 kits have been allocated.

TEAM STATUS	KIT PROVISION
Registered and paid within the first 8	Receives the organiser-provided construction kit
Registered after all 8 kits are allocated	Participates in the event without an organiser-provided kit

4. Components Provided by Organisers

The organiser-provided construction kit consists of the following items:

#	COMPONENT	TYPICAL USE IN THE ROCKET
1	Small Thermocol	Lightweight body parts, nose cone and fins
2	MDF Board	Fins, reinforcement and other flat structural parts
3	10 mL small syringes	Pneumatic components, used as specified by the organisers
4	20 mL syringes	Pneumatic components, used as specified by the organisers
5	Fevicol	Bonding Thermocol and MDF parts
6	Super Glue	Quick, strong bonding of small parts
7	Saline pipe / tubing	Tubing and connections for the pneumatic system

#	COMPONENT	TYPICAL USE IN THE ROCKET
8	Nails	Fixing parts where the organisers permit
9	Wire	Binding and reinforcement
10	Other basic miscellaneous construction materials	As specified by the organisers

STANDARDISED PROPULSION SYSTEM

The organisers will also provide the **standardised syringe-based pneumatic propulsion system** used to launch every rocket. All teams launch with the same system. No other propulsion system is part of the kit or permitted in the competition.

5. Rocket Design

Participants may independently decide the following design parameters, within the limits specified by the organisers:

DESIGN PARAMETER	TEAM'S CHOICE
Rocket body shape	Overall form and profile of the body
Nose-cone geometry	Shape and proportions of the nose cone
Number of fins	How many fins the rocket carries
Fin shape	Outline and proportions of each fin
Fin placement	Position of the fins along the body
Weight distribution	How mass is spread through the rocket
Centre of gravity	Position of the balance point
Overall rocket dimensions	Within the limits announced by the organisers
Material arrangement	How the permitted materials are combined
Aerodynamic configuration	The complete aerodynamic layout

DESIGN PRIORITIES

- Stability
- Structural integrity
- Proper centre of gravity
- Efficient aerodynamic shape
- Low drag
- Lightweight construction
- Proper centre of pressure relationship

Dimensional and weight limits, if any, will be announced by the organisers. No numerical limits apply unless officially announced.

6. Round 1 — Rocket Design & Construction

Round 1 involves the design and construction of the rocket. Teams receive the permitted materials and build their own rocket while following all dimensional, weight and safety restrictions announced by the organisers.

Participants independently determine the body configuration, nose cone, fin configuration, fin placement, weight distribution, overall geometry and material arrangement.

6.1 The objective of Round 1 is to produce a structurally sound, stable and competition-compliant rocket.

6.2 Teams must construct the rocket using the permitted materials only.

6.3 Teams are responsible for their own rocket throughout construction.

6.4 Organisers may inspect rockets at any stage of construction.

7. Technical Inspection

Before Round 2, every rocket must undergo technical inspection. The organisers and judges will verify the following:

#	CHECK	WHAT IS VERIFIED
1	Maximum permitted dimensions	The rocket is within the dimensional limits announced by the organisers
2	Rocket weight	The rocket meets the weight requirements announced by the organisers
3	Structural integrity	The rocket is soundly built and will withstand launch
4	Attachment to the launch mechanism	The rocket attaches properly to the standardised launcher
5	Compliance with competition rules	Materials and construction follow the rules

#	CHECK	WHAT IS VERIFIED
6	Safety of the design	Nothing in the design endangers participants or spectators
7	Stability-related requirements	The design meets the stability requirements for launch
8	Prohibited mechanisms	No prohibited propulsion or dangerous mechanism is present

7.1 Only rockets satisfying all required conditions will qualify for Round 2.

7.2 A rocket that fails technical inspection may be rejected from the launch round, unless the issue can be corrected within the time permitted by the organisers.

8. Round 2 — Launch & Maximum Height Challenge

Qualified rockets enter the launch round.

8.1 Each qualified team will launch its rocket using the **same standardised syringe-based pneumatic propulsion system** provided by the organisers.

8.2 Each team will launch from the designated launch area.

8.3 Launches will take place under the supervision of the event coordinators.

8.4 The primary objective is to achieve the **maximum valid flight altitude**.

8.5 The altitude measurement method will be announced by the organisers before the launch round.

8.6 The rocket must complete a valid launch according to the competition rules.

8.7 Any launch violating safety or technical requirements may be declared invalid.

The team achieving the highest **valid** altitude while complying with all competition requirements will be declared the winner.

9. Launch Procedure

STEP 1 **Rocket inspection** → STEP 2 **Team reporting** → STEP 3 **Attachment to launcher** → STEP 4 **Safety clearance**

STEP 5 **Official launch signal** → STEP 6 **Flight** → STEP 7 **Height measurement** → STEP 8 **Recording the result**

STEP	WHAT HAPPENS
1. Rocket inspection	The rocket is checked and confirmed as having passed technical inspection.
2. Team reporting	The team reports to the launch area when called by the coordinators.
3. Placement / attachment	The rocket is placed on and attached to the standardised launch mechanism under official supervision.
4. Safety clearance	Officials confirm that the launch area is clear and all participants are behind the safety boundary.
5. Official launch signal	The launch takes place only on the official signal.
6. Flight	The rocket flies freely. Nobody may assist or interfere with it.
7. Height measurement	The flight height is measured by the official measurement method.
8. Recording of the result	The result is recorded by the officials.
9. Validation of the attempt	The judges confirm whether the attempt is valid under the flight validity rules.

No participant may approach the launch area until the organisers declare it safe.

10. Flight Validity

A LAUNCH IS VALID WHEN

- The rocket is properly attached to the designated launch mechanism.
- The launch is performed only on the official signal.
- There is no unauthorised assistance.
- No prohibited propulsion is used.
- There is no interference during the launch.
- The rocket complies with all dimensional, weight and safety requirements.
- The rocket is launched from the designated area.

AN ATTEMPT MAY BE DECLARED INVALID IF

- The rocket is launched before the official signal.
- The team uses an unauthorised propulsion method.
- The rocket is intentionally assisted during launch.

- The team interferes with the measurement process.
- The rocket violates mandatory safety requirements.
- The launch is otherwise found to be unfair or non-compliant by the judges.

11. Height Measurement

11.1 The primary scoring parameter is the **maximum valid flight height**.

11.2 The organisers will specify the final measurement method before the competition.

11.3 Depending on the final event setup, the method may include an approved visual, sensor-based, video-based or other measurement system.

11.4 No specific measurement technology is guaranteed unless it is officially confirmed by the organisers.

11.5 The recorded result of the official measurement system / judging team will be considered final.

12. Scoring

The primary ranking criterion is **maximum valid altitude**. Each team is evaluated out of **100 marks** as follows.

CRITERION	WHAT IS EVALUATED	MARKS
Maximum Valid Flight Height	The highest valid altitude reached by the rocket	**70**
Rocket Stability & Flight Performance	How stable, straight and controlled the flight is	**15**
Structural Integrity	How well the rocket holds together through launch and flight	**10**
Rule Compliance & Safety	Adherence to the rules, instructions and safety requirements	**5**
Total		**100**

CRITERIA EXPLAINED

12.1 **Maximum Valid Flight Height (70):** awarded on the highest valid altitude achieved. Teams with higher valid altitudes receive higher marks. This is the main determining factor of the competition.

12.2 **Rocket Stability & Flight Performance (15):** evaluates whether the rocket leaves the launcher cleanly, flies straight and remains stable in flight.

12.3 **Structural Integrity (10):** evaluates the strength and condition of the rocket before and after launch, including the body, fins, joints and nose cone.

12.4 **Rule Compliance & Safety (5):** evaluates adherence to the rules, safety instructions and organiser directions.

Flight altitude remains the main determining factor. Aesthetic appearance and creativity do not carry marks and cannot outweigh flight performance.

13. Attempts / Re-Launch

13.1 The number of permitted attempts per team will be announced by the organisers before Round 2.

13.2 If multiple attempts are permitted, the best valid altitude will be considered for the final ranking.

13.3 Invalid attempts will not be counted as valid altitude scores.

13.4 A re-launch will be permitted only according to the official competition procedure.

14. Tie-Breaking

If two or more teams achieve the same maximum valid altitude, the tie is resolved in the order below. The team ranking higher at the first point of difference is placed higher.

COMPARE FIRST **Stability score** → COMPARE NEXT **Structural integrity** → COMPARE NEXT **Compliance & safety** → IF STILL TIED **Additional launch**

14.1 The team with the higher stability / flight-performance score ranks higher.

14.2 If still tied, the team with the higher structural-integrity score ranks higher.

14.3 If still tied, the team with the better rule-compliance / safety score ranks higher.

14.4 If the tie remains unresolved, the organisers may conduct an additional launch or tie-breaking attempt.

15. Prohibited Propulsion and Materials

The following are strictly prohibited:

- Combustion engines
- Fire-based propulsion
- Chemical propellants
- Motorised propulsion
- Pyrotechnics
- Any pressurised system other than the official standardised syringe-based pneumatic mechanism
- Fuel
- Explosives
- Compressed gas cylinders
- Electric propulsion
- Any dangerous propulsion system
- Any mechanism intended to provide an unfair launch advantage

The official syringe-based pneumatic propulsion system provided by the organisers is the **only** permitted launch mechanism.

16. Safety Rules

16.1 All participants must follow the instructions of the organisers.

16.2 Only designated officials may operate or authorise the launch mechanism.

16.3 Participants must remain behind the designated safety boundary during launch.

16.4 No participant may approach the launch area until it is declared safe.

16.5 Rockets must not contain dangerous or hazardous materials.

16.6 No sharp, explosive, burning or hazardous payloads are permitted.

16.7 Participants must not aim or intentionally launch the rocket toward people, buildings or restricted areas.

16.8 Any structurally unsafe rocket may be rejected.

16.9 A damaged rocket must be inspected before another attempt.

16.10 The organisers may immediately stop any unsafe activity.

16.11 Decisions of the safety officials are final.

17. Penalties

Violations will attract the consequences below. The severity of each penalty will be determined by the judges / organising committee.

#	VIOLATION	POSSIBLE CONSEQUENCE
1	Unauthorised propulsion	Invalid attempt or disqualification

#	VIOLATION	POSSIBLE CONSEQUENCE
2	Launching before the official signal	Invalid attempt / re-launch not permitted
3	Entering restricted areas	Score deduction or invalid attempt
4	Interfering with another team's launch	Score deduction or disqualification
5	Manipulating measurement	Invalid attempt or disqualification
6	Using prohibited materials	Invalid attempt or disqualification
7	Using unauthorised assistance	Invalid attempt or score deduction
8	Violating safety instructions	Score deduction or disqualification
9	Deliberately damaging competition equipment	Disqualification
10	Misconduct	Score deduction or disqualification

TYPES OF CONSEQUENCE

CONSEQUENCE	MEANING
Invalid attempt	The attempt does not count as a valid altitude
Score deduction	Marks are deducted from the team's score
Re-launch not permitted	The team may not re-launch after the violation
Disqualification	The team is removed from the competition

18. Disqualification

A team may be disqualified for any of the following:

18.1 Using prohibited propulsion systems.

18.2 Using dangerous materials.

18.3 Deliberately violating safety rules.

18.4 Tampering with the official launch mechanism.

18.5 Tampering with height measurement.

18.6 Cheating.

18.7 Interfering with another team.

18.8 Deliberately damaging event equipment.

18.9 Repeatedly violating instructions.

18.10 Any action that compromises the safety or fairness of the competition.

19. General Rules

19.1 Participants must report according to the announced schedule.

19.2 All teams must attend the briefing if instructed.

19.3 Teams are responsible for their own rocket during construction.

19.4 Teams must keep the construction and launch areas clean.

19.5 Participants must not enter restricted areas.

19.6 No team may interfere with another team's rocket.

19.7 Participants must maintain discipline and sportsmanship.

19.8 Organisers may inspect rockets at any stage.

19.9 Organisers may reject any rocket considered unsafe or non-compliant.

19.10 Organisers may modify the launch area or procedure if required for safety.

19.11 Any changes will be communicated before the relevant round.

19.12 Judges' decisions regarding validity, penalties and final rankings are final and binding.

20. Prize Pool

Prize Value: ₹13,000+ (including goodies)

21. Final Authority

21.1 The organisers reserve the right to interpret the rules in this rulebook.

21.2 Any situation not specifically covered in this rulebook will be decided by the organising committee / judging panel.

21.3 Decisions concerning safety, launch validity, measurement, penalties and final rankings are final.

21.4 Any rule modification will be communicated to participants before implementation.

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Best of luck to all participants!

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